

OR Endterm — Exam Outline

Endterm outline — what Exercises 1–8 will look like

Format: 120 minutes, ~120 points, **7 exercises** (SS23, SS24, SS25 all had exactly 7; the SS21 retake, now out of scope, had 8). One point $\approx$ one minute. Aids: non-programmable calculator, analog dictionary, ruler. No cheat sheet.

Below: 8 slots. Slots 1–7 are the reliable skeleton. Slot 8 is the rotating one — in a 7-exercise paper it gets folded into slot 6 or 7, or replaces slot 1.

What actually appeared

The SS21 retake is **out of scope** — too old to be representative — so the four papers below are the evidence base.

#	SS21 endterm	SS23	SS24	SS25
1	Sensitivity (16)	Duality (19)	Multiple choice (14)	Multiple choice (16)
2	Duality (6)	IP modelling (22)	Duality (16)	Simplex/sensitivity (16)
3	IP modelling (35)	Branch & Bound (16)	Branch & Bound (16)	Duality (15)
4	Branch & Bound (13)	Knapsack DP (14)	IP modelling (22)	IP modelling (21)
5	Flow + TU (14)	Matroids (14)	Matroids (14)	Branch & Bound (14)
6	Convex functions (13)	TSP heuristic (12)	IP + NP-hardness (18)	Max-flow/min-cut (14)
7	Column generation (23)	Critical points (23)	Nonlinear (20)	Convexity + KKT (20)

Never absent: IP modelling, branch & bound, duality, nonlinear. **Usually there:** multiple choice (since SS24), simplex/sensitivity, one combinatorial topic. **Rare:** column generation (SS21 only, but 23 points when it came).

All four papers have **exactly 7 exercises**, which strengthens the 7-slot reading below.

Exercise 1 — Multiple choice · 14–16 pts

Type. 7–8 items, 2 pts each, "mark all correct answers (there can be more than one)". **Partial credit with a penalty:** all correct $\rightarrow$ 2 pts; exactly one error $\rightarrow$ 1 pt; more $\rightarrow$ 0 pts. Selecting nothing = 0. So **never leave one blank** — guessing dominates.

Topics. One item per lecture block: duality, Christofides/TSP, cutting planes, sensitivity, matroids, total unimodularity, simplex cycling, gradient descent / KKT.

Example (SS25 1a):

Which of the following statements are correct?

- ☐ Every linear program has a dual program.
- ☐ If a primal constraint is not tight, then its dual variable is zero.
- ☐ If the primal is infeasible, then the dual is unbounded.
- ☐ If the primal is unbounded, then the dual is infeasible.

→ 1st ✓ (always), 2nd ✓ (complementary slackness), 3rd ✗ (infeasible primal $\Rightarrow$ dual is unbounded **or** infeasible), 4th ✓ (weak duality).

Watch. The traps are always the "always/never" quantifiers and the 3×3 primal–dual outcome table.
Practice: SS24 P1, SS25 E1.

Exercise 2 — Simplex, tableau forensics, sensitivity · 14–19 pts

Type. You're handed an optimal (or corrupted) tableau and asked to read things off it. Almost never "run simplex from scratch" — it's *reverse engineering*, then a sensitivity chain of 3–4 sub-questions.

Topics. Row 0 = $c_B B^{-1}A - c$, reduced costs, shadow prices, cost ranging, RHS ranging, degeneracy, pricing a new column.

Example (SS25 E2, "Lemonade or Iced Tea"). Given a feasible tableau with **wrong Row 0**:

Basis	x_1	x_2	s_1	s_2	s_3	s_4	RHS
Row 0	0	-1	2	1	0	0	0
x_1	1	0	1/3	-1/3	0	0	20
x_2	0	1	-1/6	2/3	0	0	10

a) What is the correct Row 0? b) Is the corrected tableau optimal? Why? c) Range of the Lemonade price p_1 keeping this basis optimal. d) Which perturbation of the water capacity makes the BFS degenerate — which basic variable hits zero first? e) New drink uses 3 L water + 1 kg lemon: write its reduced cost and the condition on p_S for it to enter the basis.

Watch. The full sensitivity chain — shadow price $\rightarrow$ validity range $\rightarrow$ violate the range $\rightarrow$ recompute the new shadow price after a basis change.

Exercise 3 — Duality and complementary slackness · 6–19 pts

Type. Three parts, near-verbatim every year: (a) derive the dual, (b) state both sets of CS conditions, (c) use CS to **disprove** optimality of a given point.

Topics. SOB table ($\max/\min \times \leq/\geq/= \times \text{free}/\geq 0$), weak & strong duality, the 3×3 outcome table, occasionally a zero-sum game.

Example (SS25 E3):

$$\begin{aligned}
 \text{(P)} \quad & \max 3x_1 + 2x_2 + x_3 \\
 \text{s.t.} \quad & x_1 + 2x_2 + x_3 \leq 10 \\
 & 2x_1 - x_2 + 2x_3 \geq 4 \\
 & x_1 + x_2 - x_3 = 3 \\
 & x_1, x_2, x_3 \geq 0
 \end{aligned}$$

a) Derive (D). $\rightarrow y_1 \geq 0, y_2 \leq 0, y_3$ free, three $\geq$ dual constraints. b) Primal CS: $x_j (A^T y - c)_j = 0$. Dual CS: $y_i (Ax - b)_i = 0$. c) Show $x = (2, 2, 1)^T$ is **not** optimal — build the dual system CS forces, show it has no feasible solution. You never solve the LP.

Watch. Part (c) is worth the most and is pure procedure. Also: "is (D) unbounded, infeasible, or finite — without solving it".

Exercise 4 — IP modelling · 21–35 pts

The biggest single block on every paper. A word problem, 5–7 sub-parts, each asking you to encode one English sentence as linear (in)equalities. Prompt is verbatim each year:

"Formulate the following constraints as linear (in)equalities. You may introduce additional variables. If you do so, please also state their intuitive meaning in words."

Topics. Assignment, capacity, logical implication, big-M, either/or, at-most-k, conditional counting, multi-objective scalarisation.

Example (SS25 E4, "School Planning"). $y_i \in \{0, 1\}$ school i built, $x_{\{i, j\}} \in \{0, 1\}$ student $j \rightarrow$ school i :

Sub-part	English	Answer
a	each student to exactly one school	$\sum_i x_{\{i, j\}} = 1 \forall j$
b	capacity	$\sum_j x_{\{i, j\}} \leq c_i y_i \forall i$
c	assign only if built	$x_{\{i, j\}} \leq y_i \forall i, j$
d	built schools ≥ 5 km apart	$y_i + y_{i'} \leq 1 \forall i \neq i' \text{ with } d_{\{i, i'\}} < 5$
e	>200 students in zone A $\Rightarrow \geq 300$ in zone B	introduce $z \in \{0, 1\}$, two big-M rows
f	one objective: min cost, max preference	$\min \sum_i f_i y_i - \lambda \sum_{\{i, j\}} s_{\{i, j\}} x_{\{i, j\}}$

Part (e) is the graded one — the implication pattern:

$$\begin{aligned}
 \Sigma(\text{zone A}) &\leq 200 + M \cdot z & (z = 1 \text{ forced if more than } 200) \\
 \Sigma(\text{zone B}) &\geq 300 \cdot z & (\text{obligation activates only when } z = 1)
 \end{aligned}$$

Watch. Always name your helper variables in words — undocumented variables score zero. Past scenarios: beer gardens (SS21), ice cream (SS23), Tour d'Allemagne stages (SS24), schools (SS25).

Exercise 5 — Branch and Bound · 13–16 pts

Type. Two flavours. (i) Classic: solve a 2-variable IP by B&B, subproblems solved graphically on the plot provided. (ii) SS25's variant: the tree is **ink-blotted** and you reconstruct it from a table of candidate subproblems.

Topics. LP relaxation as bound, branching rule, the three pruning rules (integrality / bound / infeasibility — and their direction flips between max and min), FIFO vs LIFO order, sometimes a Gomory cut.

Example (SS25 E5, "Gondolf the wizard"):

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max 3x1 + 2x2
s.t. 2.4x1 + x2 ≤ 9.15
      x1 + x2 ≤ 6
      x1, x2 ∈ ℕ0

P0: x = (2.25, 3.75), Z0 = 14.25, branch on x1
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a) Match the blotted nodes P_1, P_2, P_4, P_5, P_6 to rows of the given subproblem table. b) Recover the missing branching constraints. c) Optimal integer solution and objective value. d) In what order would the nodes be explored under (i) FIFO and (ii) LIFO, stopping as soon as optimality is confirmed?

Watch. Part (d) is free points if you know the two traversal orders cold. Pruning a node whose bound merely *ties* the incumbent is allowed — say so explicitly.

Exercise 6 — Combinatorial block · 12–18 pts

Type. Rotates between five topics. This is where "show or disprove / give a counterexample" lives.

Topic	Year	Shape
Matroids	SS23, SS24	verify the 3 axioms, or give a set system that violates one
Network flow / min cut	SS21, SS25	Ford–Fulkerson, max-flow = min-cut, counterexamples
Knapsack DP	SS23	fill the DP table, $O(nW)$ vs $O(nV)$
Total unimodularity	SS21	check the 3 sufficient conditions on an incidence matrix
TSP / approximation	SS23, SS24	disprove a heuristic, or run MST-doubling

Example A (SS25 E6, max-flow counterexamples). "Use the given graph to find a counterexample and describe why it refutes the statement."

a) If all capacities are odd, there is a maximal s – t flow f with $f(e)$ odd for all e . b) Adding $\lambda \in \mathbb{N}$ to all capacities does not change the minimal cuts.

Both are false; you draw capacities on the given diamond graph $s \rightarrow a, b \rightarrow c \rightarrow t$ and exhibit them. (b) fails because adding λ penalises cuts with *more* edges — a 2-edge cut and a 1-edge cut that tie will separate once you shift everything up.

Example B (SS23 E4, knapsack DP). 5 vinyl records, capacity 5 kg. $V_{\max} = 12 > W_{\max} = 5$, so break down **by weight** ($O(nW)$ is the smaller bound). Fill the 6×6 table, read off profit 6 from the corner, backtrack for the item set.

Example C (SS24 P6). Min number of ATMs covering every street $\rightarrow \min \sum_i x_i$ s.t. $x_i + x_j \geq 1$ $\forall (i,j) \in E$. Then: "which NP-hard problem is this?" $\rightarrow$ vertex cover. Then: "how many for a complete graph on n nodes?" $\rightarrow n - 1$.

Watch. For TU, reproduce the three numbered sufficient conditions — naming "Ghouila-Houri" earns nothing, the lecture never names it. For TSP, the papers mislabel MST-doubling as "Christofides"; execute MST-doubling if the wording matches theirs.

Exercise 7 — Nonlinear optimization · 12–23 pts

Type. Two flavours, and some years you get both (that's what makes it slot 7 + 8).

7a — Unconstrained. Compute ∇f , solve $\nabla f = 0$, classify each critical point via the Hessian (leading principal minors, eigenvalues as fallback). Frequently parameterised.

Example (SS23 E7): for $\alpha > 0$, classify all critical points of $f_\alpha(x,y) = -\frac{1}{4}x^4 - \frac{1}{4}y^4 + \alpha xy$ as a function of α . You get a case split on α , not a single answer.

Example (SS24 P7a): $f(x,y) = x^2 + y^2 + e^{2x} - (8x + 6y) + 2xy$. Trace and determinant of $\nabla^2 f$ are both positive everywhere $\Rightarrow$ strictly convex $\Rightarrow$ at most one minimum, no maxima.

7b — Constrained: convexity, Slater, KKT. A modelling story leading to a convex program, then the four KKT blocks.

Example (SS25 E7, "Minimal Circle Enclosure" — smallest circle covering points p_i):

- a) Define a convex function. Prove any norm $\|\cdot\|$ on $\mathbb{R}^d$ is convex.
- b) Write the problem as $\min_{\{r,c_1,c_2\}} r$
s.t. $\|p_i - c\|_2 - r \leq 0 \quad \forall i, \quad r \geq 0$
Show f and every g_i are convex.
- c) Show Slater's condition holds — give an explicit strictly feasible (r,c_1,c_2) .
- d) From $L = f + \sum \lambda_i g_i - \mu r$, use KKT to derive $\sum \lambda_i = 1, \quad c^* = \sum \lambda_i p_i$.
- e) What does $\lambda_i > 0$ imply about p_i ? $\rightarrow p_i$ lies ON the circle (CS: its constraint is tight).

Watch. Part (a) — proving convexity straight from the definition — is worth real points and people skip it. Know the split: **Slater $\Rightarrow$ KKT point is a global optimum; LICQ $\Rightarrow$ KKT gives candidates only.**

Exercise 8 — The wildcard · 12–23 pts

Every in-scope paper runs to 7 exercises, so treat this slot as unlikely. Historically only one thing has filled it:

Column generation (SS21 A7, 23 pts). Set up the restricted master problem, count the columns of A , formulate the pricing problem, argue termination.

Example: m friends with luggage w_i , cars of capacity W , some pairs refuse to share. Columns of A = valid fill patterns. "How many columns does A have if every pair is feuding?" $\rightarrow$ exactly m (each person alone). "When is column generation especially worthwhile?" $\rightarrow$ when $|S|$ is small, so the pattern count explodes.

That question is **CE demo D7.2 word-for-word**, so if it ever reappears the demo is the prep. Otherwise this slot merges upward into 6 or 7.

How to spend the 120 minutes

E4	IP modelling	~22 pts / 25 min	← largest, do it while fresh
E7	Nonlinear	~20 pts / 22 min	← second largest
E1	Multiple choice	~15 pts / 12 min	← best pts/min, never blank
E2	Simplex/sens	~16 pts / 18 min	
E3	Duality	~15 pts / 15 min	
E5	Branch & Bound	~14 pts / 14 min	
E6	Combinatorial	~14 pts / 14 min	

Scan all problems first, start with your strongest. **Document every approach** — an undocumented answer scores zero even when it's right.